UNISWAP V3 · UNISWAP V4 · LP NFT COLLATERAL · ERC-4626 PAIRVAULT · FEETO · LIQUIDATIONFEEBPS · CHAINLINK FEEDS · CRE KEEPER

LPFY

Uniswap LP NFT lending on LPFY

Borrow Circle USDC against **Uniswap V3 and V4** concentrated-liquidity position NFTs. Each pair has an isolated ERC-4626 PairVault for idle USDC. Adapters value and unwind those positions under an LTV cap; liquidations repay the vault, then split surplus between `feeTo` and the borrower. Chainlink feeds and an optional CRE keeper support pricing and automation in the background.

Overview · Uniswap collateral · ERC-4626 vaults · Flows · Share pricing · Fee structure · Liquidation · Roles · References · Deployed

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1. Overview

UNISWAP

V3 / V4 LP NFT collateral

Core use case: lock a Uniswap concentrated-liquidity position NFT (V3 NonfungiblePositionManager or V4 PositionManager) and borrow Circle USDC from the matching pair under LTV limits.

LENDERS

ERC-4626 PairVault

One vault per pair. Supply with `deposit` / `redeem`. Idle USDC sits in the vault, not on the market.

FEES

Surplus on liquidate

`liquidationFeeBps` of post-debt surplus → `feeTo` (separate from Ownable owner). Rest → borrower.

> LPFY is built around **Uniswap LP NFTs as borrow collateral** — V3 and V4 adapters value positions, enforce pair matching, and unwind liquidity on liquidation.

2. Uniswap V3 & V4 collateral

Borrowers post Uniswap LP NFTs. The market routes each position through a version-specific adapter so V3 and V4 can share the same lending book without mixing position formats.

> **What we use from Uniswap (on-chain AMM — no SDK / API):**
>
> - **Uniswap V3 AMM** — NonfungiblePositionManager (LP NFT custody / unwind), Factory + Pool `slot0` (valuation)
> - **Uniswap V4 AMM** — PositionManager (LP NFT custody / unwind), StateView (ticks / liquidity reads); native ETH mapped to WETH
> - **SwapRouter02** — `exactInputSingle` during liquidation to swap non-USDC legs → Circle USDC (with fee-tier fallback)
> - **Not used:** Uniswap SDK, Trading API, Universal Router, CCA, or custom v4 hooks

V3

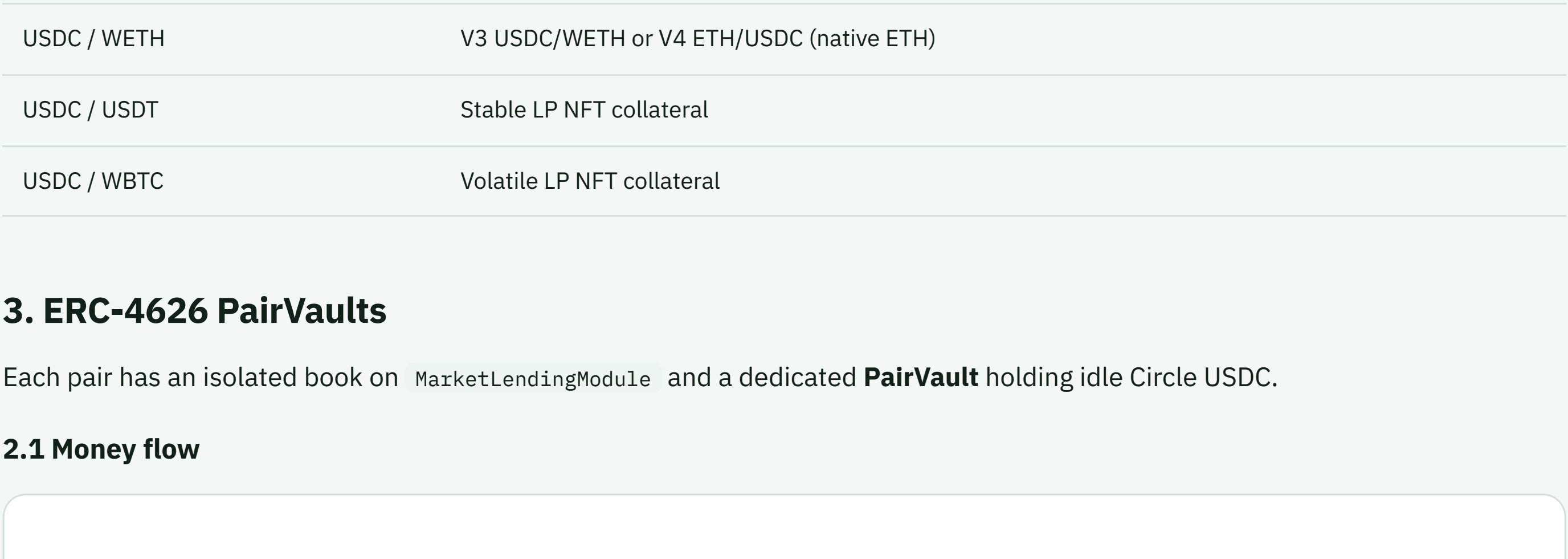
V3Adapter

Reads and unwinds Uniswap V3 positions via the NonfungiblePositionManager. Collects fees/tokens, then the market swaps non-USDC legs to Circle USDC on liquidation.

V4

V4Adapter

Same borrow / liquidate path for Uniswap V4 PositionManager NFTs (including native-ETH pools where configured). StateView supports valuation reads.



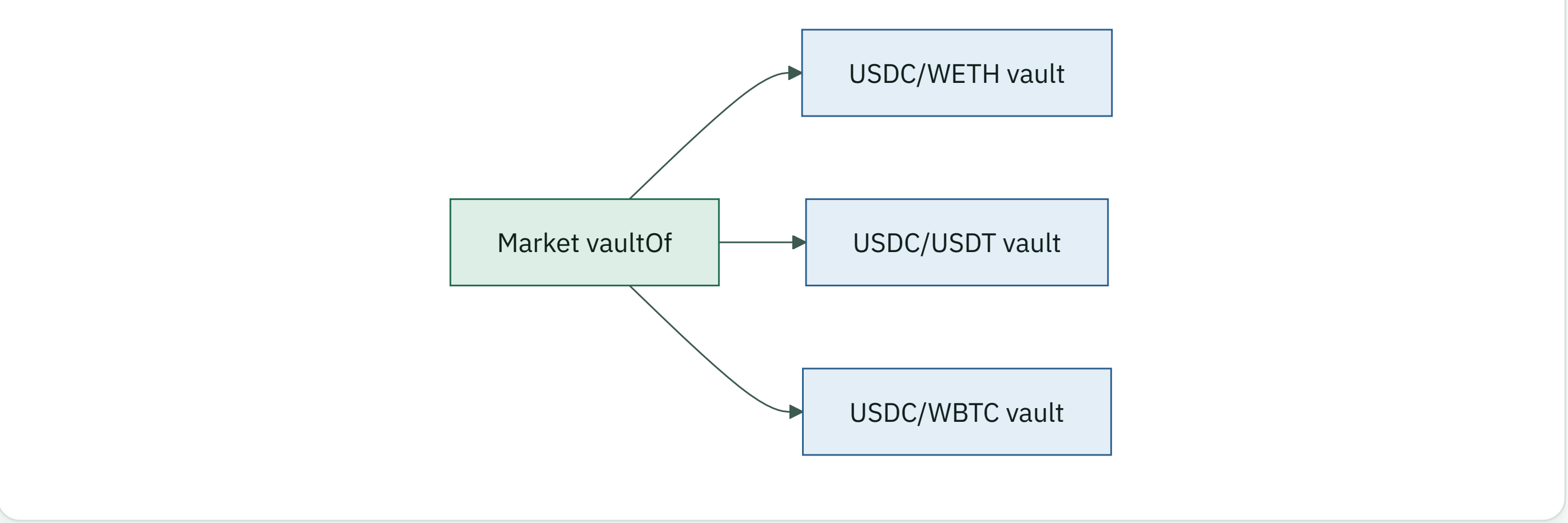
One market entrypoint; adapters isolate Uniswap V3 vs V4 position mechanics.

PAIR (SEPOLIA)	TYPICAL UNISWAP USE
USDC / WETH	V3 USDC/WETH or V4 ETH/USDC (native ETH)
USDC / USDT	Stable LP NFT collateral
USDC / WBTC	Volatile LP NFT collateral

3. ERC-4626 PairVaults

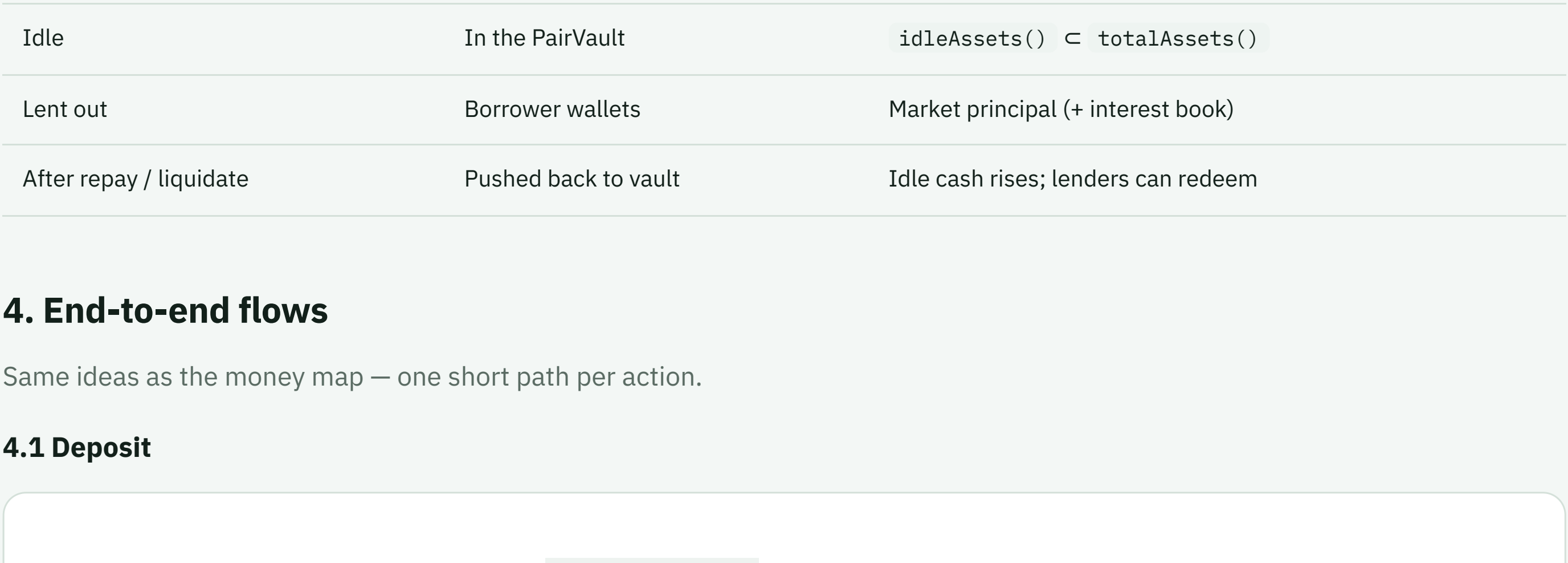
Each pair has an isolated book on `MarketLendingModule` and a dedicated **PairVault** holding idle Circle USDC.

2.1 Money flow



Lender talks only to the vault. Market moves liquidity for borrowers — it does not hold idle lender cash.

2.2 Isolation



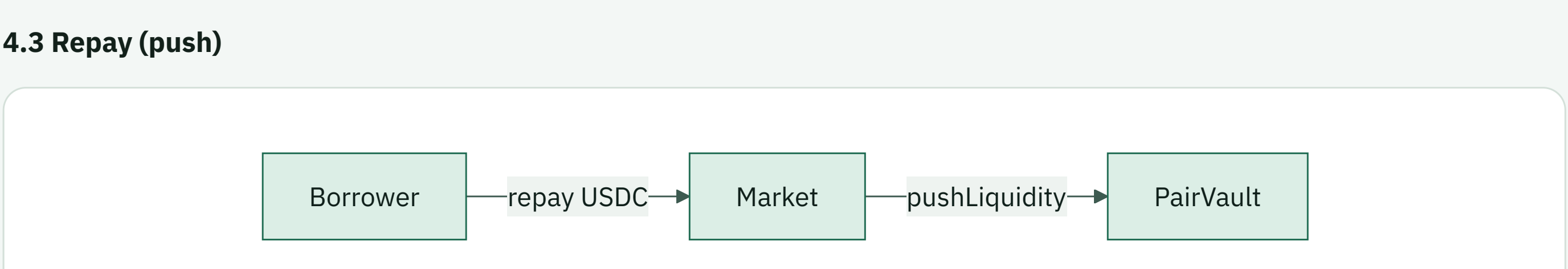
Liquidity never mixes across pairs.

STATE	WHERE USDC SITS	COUNTED IN
Idle	In the PairVault	`idleAssets()` ⊂ `totalAssets()`
Lent out	Borrower wallets	Market principal (+ interest book)
After repay / liquidate	Pushed back to vault	Idle cash rises; lenders can redeem

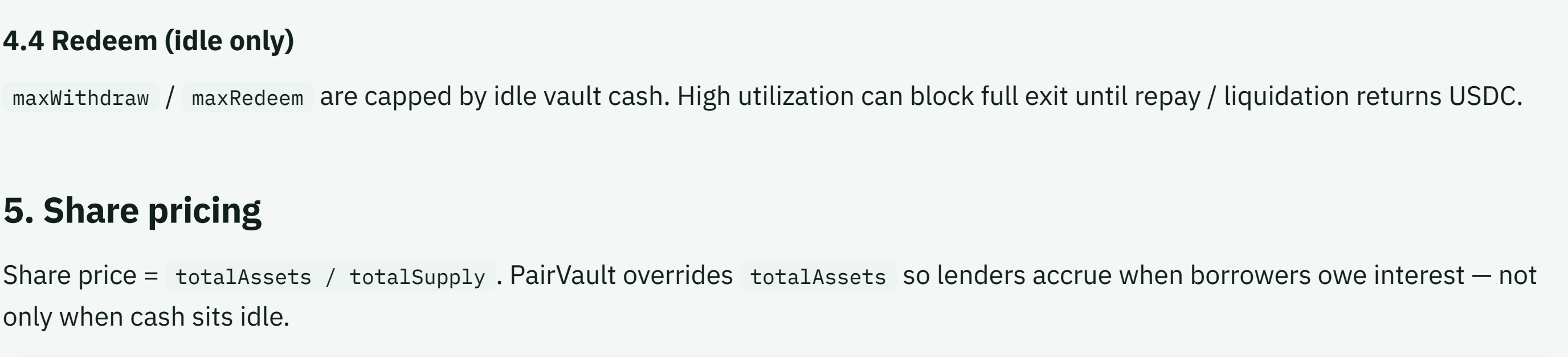
4. End-to-end flows

Same ideas as the money map — one short path per action.

4.1 Deposit

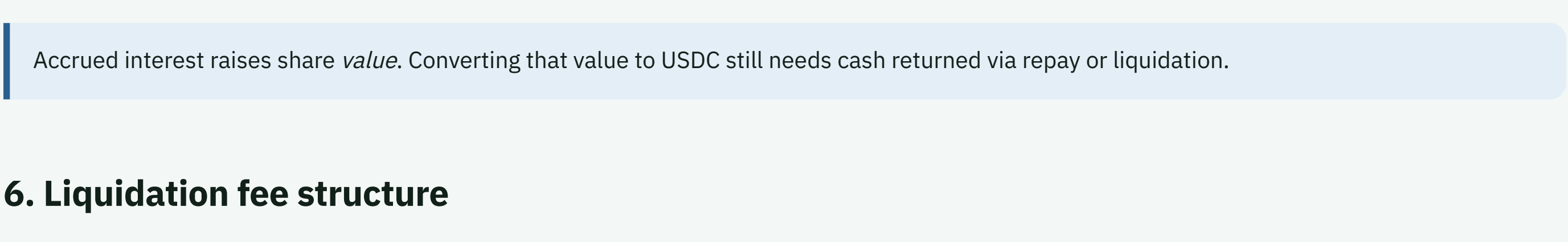


4.2 Borrow against Uniswap LP NFT



Collateral is always a Uniswap V3 or V4 LP NFT for the enabled pair.

4.3 Repay (push)



4.4 Redeem (idle only)

`maxWithdraw` / `maxRedeem` are capped by idle vault cash. High utilization can block full exit until repay / liquidation returns USDC.

5. Share pricing

Share price = `totalAssets / totalSupply`. PairVault overrides `totalAssets` so lenders accrue when borrowers owe interest — not only when cash sits idle.

```
totalAssets() = idleAssets() + assets owed by the market
              = USDC idle in vault
              + totalPrincipal + totalAccruedInterest
```

> Accrued interest raises share *value*. Converting that value to USDC still needs cash returned via repay or liquidation.

6. Liquidation fee structure

Fees apply only on successful liquidation when recovery exceeds debt. They are configured on the market (not on the vault or any external keeper).

VARIABLE	MEANING
`liquidationFeeBps`	% of **surplus** (after vault debt repaid), in basis points. `1000` = **10%**
`feeTo`	Wallet that receives the protocol fee — **independent of Ownable `owner`**
Borrower surplus	Remaining surplus after the fee
Shortfall	If recovery < debt → no fee, no surplus to borrower

> **Admin (owner only):** `setLiquidationFeeBps` · `setFeeTo`

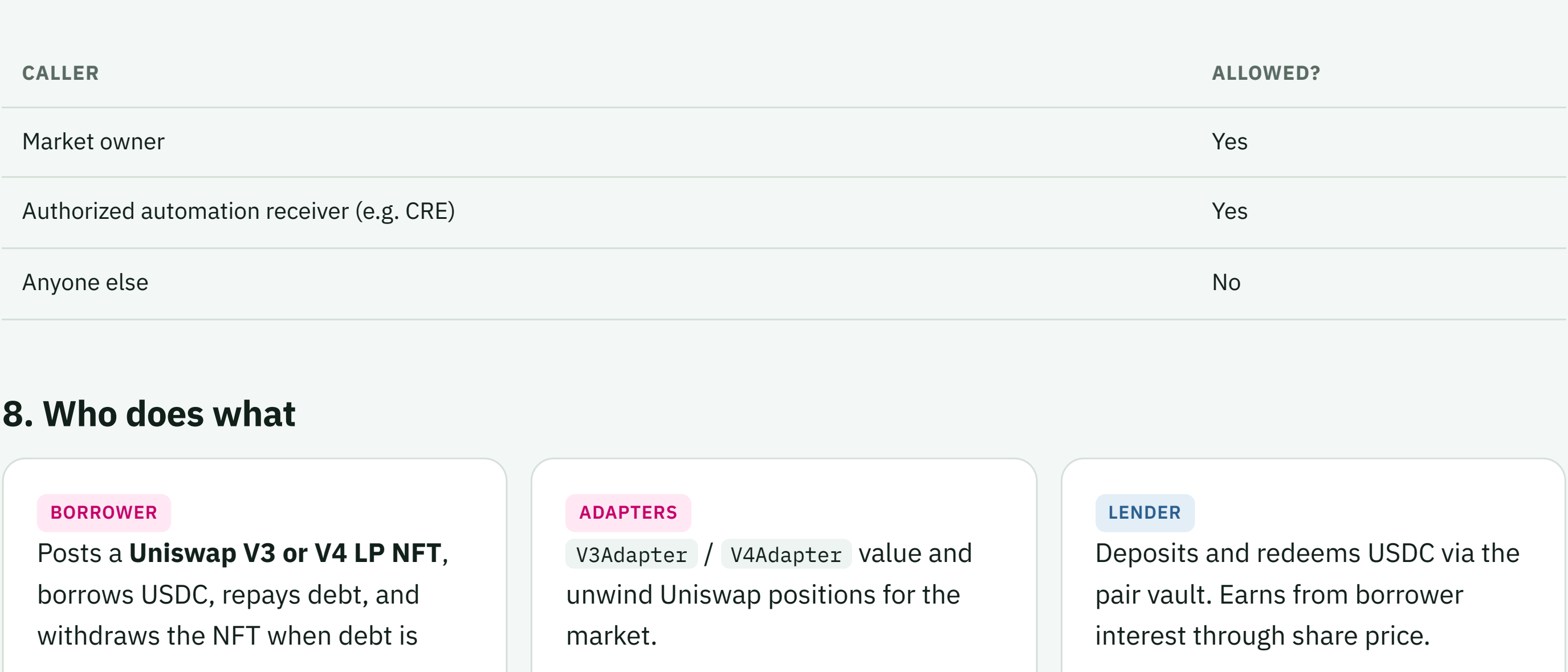
Surplus split

```
debt repaid to PairVault = principal + interest
surplus = recoveryUSDC − debt
fee = surplus × liquidationFeeBps / 10000 → feeTo
borrower gets = surplus − fee
```

7. Liquidation path

When a loan is underwater, the Uniswap LP NFT is unwound through the matching V3 or V4 adapter, tokens are swapped to USDC, and the pair vault is repaid first.

1. Loan is underwater (debt / collateral ≥ liquidation threshold).
2. Authorized caller invokes `liquidate` on the market.
3. **Uniswap adapter** unwinds the LP NFT → collect underlying tokens (empty NFT returned to borrower).
4. Non-USDC legs are swapped to Circle USDC.
5. Debt is pushed back into the pair’s PairVault.
6. `liquidationFeeBps` of remaining surplus → **feeTo**.
7. Leftover surplus → borrower; loan closed.



Anyone may preview a liquidation. Execution is restricted to the market owner (and optionally an authorized automation receiver).

CALLER	ALLOWED?
Market owner	Yes
Authorized automation receiver (e.g. CRE)	Yes
Anyone else	No

8. Who does what

BORROWER

Posts a **Uniswap V3 or V4 LP NFT**, borrows USDC, repays debt, and withdraws the NFT when debt is zero.

ADAPTERS

`V3Adapter` / `V4Adapter` value and unwind Uniswap positions for the market.

LENDER

Deposits and redeems USDC via the pair vault. Earns from borrower interest through share price.

OWNER

Configures risk params, fees, and may liquidate. Liquidate and admin actions are owner-only in the app.

FEETO

Receives the protocol cut of liquidation surplus when recovery exceeds debt.

SUPPORTING

Price feeds (Chainlink) for LTV checks; optional CRE keeper to auto-call `liquidate`. Neither defines the Uniswap collateral model.

9. References

GIT

Repository

github.com/unblocktechie/LPFy

Key packages: `smart-contracts/` · `web/` · `docs/` · `cre/` (optional keeper)

FILES

Where to look in the repo

PATH	ROLE
`smart-contracts/contracts/markets/MarketLendingModule.sol`	Loans, LTV, Uniswap collateral, liquidate, fees
`smart-contracts/contracts/adapters/V3Adapter.sol`	Uniswap V3 LP NFT value / unwind
`smart-contracts/contracts/adapters/V4Adapter.sol`	Uniswap V4 LP NFT value / unwind
`smart-contracts/contracts/markets/PairVault.sol`	ERC-4626 idle USDC vault
`smart-contracts/deployments/sepolia-markets.json`	Live Sepolia addresses + `webEnv`
`web/src/features/supply/SupplyPanel.tsx`	Vault `deposit` / redeem UI
`smart-contracts/contracts/markets/CreLiquidationReceiver.sol`	Optional automation → `liquidate`
`cre/liquidation-keeper/workflow.ts`	Optional CRE keeper

Quick recap

- **Uniswap-first:** borrow USDC against V3 or V4 LP NFTs via dedicated adapters.
- **Isolated pools:** each pair has its own books and PairVault.
- **ERC-4626:** lenders use standard deposit / mint / withdraw / redeem.
- **Custody:** idle USDC in vault; market pulls on borrow, pushes on repay / liquidate.
- **Pricing:** `totalAssets = idle + principal + accrued interest`.
- **Fees:** after vault is repaid, `liquidationFeeBps` of surplus → `feeTo`; rest → borrower.
- **Liquidation:** unwind Uniswap position → swap → repay vault; owner (or optional keeper) may execute.

Deployed contracts (Sepolia)

Canonical file: `smart-contracts/deployments/sepolia-markets.json`. Full setup: SETUP_AND_DEPLOY.md.

CONTRACT	ADDRESS
`MarketLendingModule`	`0x618a17C71b1D3922172E1b77D6e3D17704DE9f41`
`ValuationOracle`	`0x16E03515cf524B5D0D0db6E44F342ef0af7301e3`
`V3Adapter`	`0x48ee590E073f5ae2FDFCEA465e6281115308dDc9`
`V4Adapter`	`0x646DB0572422424947262fC3Bda2940f51504759`
`BorrowRateConfig`	`0x96ff2ce7B8AEfEadCAF460f41af613e7640AE20B`
`StaticApySource`	`0x02874283bd490350Ace341bc9Cdd4E7A686CB4E1`
`CreLiquidationReceiver`	`0x32866b81408d0260259BC0109b9F0a300Cdca132`
`FeeFallbackSwapRouter`	`0xEB66D88296C1694d79D578A700cb4EE0d6dC8baf`
`PairVault` USDC/WETH	`0xE47ABe4b88Fa2F47f4913710e8ecfBDE1205A5dc`
`PairVault` USDC/USDT	`0x9cf1CF796147DD6c4Ff2f57E91F67fe14397d8eF`
`PairVault` USDC/WBTC	`0xE374DEF41E1c63d54FFeB7B4a3fbA554574A4F84`

LPFY Markets · Uniswap V3/V4 LP NFT lending · GitHub · Contract architecture · Chainlink CRE (optional)